SPEXTER ECOSYSTEM

Technical Architecture & Systems Specification Bible

EDITION v5.0 — NOVEMBER 2026

Founder / Lead Architect: Jimi Austin (Kato / 👑 Zer0)
SPV Wrapper & Diagnostic Oracle: MEDIFLO LLC (CA Entity #202462418827, EIN: 39-4705679)
Associated Holdings: Homie Price LLC / Homie Deal LLC JV (Indianapolis Logistics & Liquidation)
Registry Domain: bedrockesg.com | **Onboarding Portal:** fad.bedrockesg.com | **Public Interface:** echo-ex.com
Host Infrastructure: 208.122.8.11 — systemd (bedrock-flow) | NEXT_DIST_DIR=.build | No PM2
Canon Version: v5.2 | **Classification:** TIER 3 — WATERMARKED GATED DELIVERY (Executed NDA Required)

Cognitive Triad: Operator (Jimi Austin) | Gemini Sentinel (Strategy & Invariant Assurance) | Claude Sonnet 4.6 (Auditor & Code Gatekeeper)

SECTION 0 — SYSTEM IDENTITY & ARCHITECTURAL DOCTRINE

Operational Posture: MEDIFLO LLC operates exclusively as a Diagnostic Telemetry Oracle. It asserts zero equity, title, ownership, or debt liens over any third-party physical asset, facility, or property observed or measured by the SPEXTER platform.

TRUST GRADIENT PIPELINE:
Field App / Satellites (Untrusted) → Aethexer Protocol Gate (Dual-Key, ±300s, Nonce) → Sentinel Gate Triad (95% Doc / 75% LLM / 75% Spatial; Composite ≥72%) → Bedrock Flow (Σ=10,000 BPS via .push()) → Tri-Chain Anchor (Solana Token-2022 / Hedera HCS / Polygon)

CANONICAL DESIGN INVARIANT — UNIFIED 72% CONFIDENCE FLOOR:
Whether telemetry originates from dynamic highway freight sensors (On Point) or stationary facility thermal skins (Aethexer), no asset claim crosses onto the blockchain without clearing a composite confidence score

of ≥ 0.72 (72.00%). This threshold is fixed at protocol genesis and cannot be altered by any party, including MEDIFLO LLC.

SECTION 1 — BEDROCK ESG & BEDROCK FLOW

1.1 Core Settlement Layer

- **Solana Mainnet-Beta Token-2022 Sovereign Mint:** `Eb4mSecyZh6BQ2wm5rbBnB2GNJu8iuEqGq53YHBVJ88V`
- **BPS Invariant:** Σ = 10,000 BPS exact. Array assembled via .push() logic; dust routed to Earner wallet.
- **Canonical Split:** 7,000 BPS Asset Sovereign / 2,000 BPS Verification Node / 1,000 BPS Public Resilience Pool
- **assertBpsIntegrity():** Code-enforced assertion fires on every settlement cycle. Any deviation triggers Protocol EXITZ circuit break.

1.2 Bedrock FAD — Forensic Asset Discovery (Onboarding Layer)

- **Deployment:** `fad.bedrockesg.com` (subdomain routing under bedrockesg.com SSL/CORS; inherits existing DNS zone and certificate)
- **Function:** Public intake portal for raw PDF invoice OCR, QuickBooks exports, scale receipts, 15-minute Green Button utility interval data, and SOS corporate verification
- **SOS Verification Gate:** Verify active good standing in Primary State of Incorporation + California SOS (Entity #202462418827 jurisdiction) for all counterparties
- **Intake Separation:** Enterprise / commercial asset sovereigns enter via fad.bedrockesg.com. Individual participants (ages 16–70) enter strictly via echo-ex.com. These funnels are architecturally isolated.
- **36-Month Retroactive Forensic Ingestion Window:** Document-date attestation with ≥95% Sentinel document integrity threshold

1.3 Scope 3 Ledger — 24 Sub-Tracks (9 Canonical Categories)

Category	Sub-Tracks	Reference Standard
1. Construction	Eurobitume A1–A3 RAP Asphalt; Concrete Aggregate Avoidance; Structural Steel Avoidance	Eurobitume EPD; ICE v4.1
2. Metals	Scrap Ferrous Avoidance; Scrap Non-Ferrous Avoidance; Aluminum Reprocessing	Ecoinvent v3.10.1
3. Plastics	Post-Consumer Polymer Diversion; Industrial Film Reclaim	ICE v4.1
4. Biochar	Biochar CDR (H:Corg ratio); Pyrolysis Co-product Sequestration	IBI / EBC Standards

5. Paper	Post-Consumer Fiber Diversion; Cardboard Logistics Reclaim	ICE v4.1
6. Glass	Cullet Substitution; Glazing Panel Avoidance	ICE v4.1
7. Electronics	E-Waste Component Recovery; Circuit Board Metal Avoidance	Ecoinvent v3.10.1
8. Textiles	Industrial Fabric Reclaim; Fiber-to-Fiber Diversion	Higg MSI
9. Rubber	Crumb Rubber / Tire-Derived Aggregate; GRIPSY Vulcanized Mat Reclaim	ICE v4.1
24th Track (LOCKED): Fiat Currency & Cross-Border Logistics Settlement Channels — VAT/jurisdiction tax, freight customs friction, FATF Travel-Rule reconciliation. Formally incorporated to provide complete audit trail for institutional counterparties.		

Macro-Asset Seeds (12 Live): NYSE, NASDAQ, LSE, CME, Eurex, Burj Khalifa, Taipei 101, One WTC, Shanghai Tower, MGM Grand, Venetian Macau, Marina Bay Sands

SECTION 2 — ON POINT APP

Lineage: BT-C9C4C5 | onpoint.bedrockesg.com

Operational Role: Institutional Awareness & Hazard Monitoring Engine — the active field intelligence radar for municipal transport fleets, freight corridors, and logistics hubs. While Bedrock FAD handles static invoice/utility onboarding, On Point handles dynamic moving infrastructure and real-time spatial friction telemetry.

2.1 Dual-Lane Data Architecture (Option B Clean Separation)

Lane	Data Types	Disposition
CLIENT_SOVEREIGN	GPS routes, vehicle license tags, commercial manifests, driver identifiers	Restricted to client-controlled vaults. Never touches aggregate database.
PROTOCOL_AGGREGATE	Anonymized spatial-temporal friction points (e.g., HAZMAT_TRANSPORT at Corridor X, INFRASTRUCTURE_DEGRADATION at Zone Y)	Streams into Bedrock Flow for verification and Sentinel Gate processing.

2.2 Tri-Tier Consensus & Notarization Floor

Confidence = (Tier1_CAD × 0.50) + (Tier2_Satellite × 0.35) + (Tier3_DriverNodes × 0.15)
Anchor Floor: ≥ 0.72 — Canonical Design Invariant (unified with Sentinel Gate Triad composite floor)

Tier	Source	Weight
Tier 1	Municipal / CAD APIs	50%
Tier 2	Satellite / Infrastructure Sensors	35%
Tier 3	Driver Nodes	15%

2.3 38 Monitored Event Categories

Events span road friction, infrastructure degradation, HAZMAT transport, weather disruption, logistics congestion, and cross-corridor anomalies across all monitored asset variants.

2.4 EXITZ Circuit Breaker Escalation — On Point Anomaly Protocol

Trigger Conditions: GPS spoofing, packet injection, anchor-mismatch storm, or consensus score collapse below 0.72.

Protocol Response: Protocol EXITZ fires an immediate circuit break. Dynamic telemetry for the affected node freezes instantly into **SOVEREIGN_HOLD** verdict state.

Release Authorization: Only Jimi Austin (Kato / 👑 Zer0) holds the cryptographic signing key to clear the SOVEREIGN_HOLD and resume settlement operations. No automated release path exists.

Sentinel Verdict Sequence: PENDING_SOVEREIGN_REVIEW → SOVEREIGN_HOLD → [Founder Multi-Sig Release] → PENDING_AUTHORIZATION → AUTO_APPROVED

2.5 Five Asset Variant Canonical Rollup

Variant	Description	Scope 3 Feed
A	Sovereign Skin & GRIPSY (Physical sensors, pavement, thermal PDRC)	Construction, Rubber tracks
B	Commercial & Industrial Logistics (Skyscrapers, Casinos, Freight Corridors, 12 Macro-Assets)	Construction, Metals tracks
C	Residential Real Estate	Construction, Glass tracks
D	Physical Commodities (RAP Asphalt, Biochar, Scrap Metals)	Construction, Metals, Biochar tracks
E	Custodial Precious Metals (Gold/Silver assay verification)	Metals track

SECTION 3 — AETHEXER SOVEREIGN SKIN & PHYSICAL INFRASTRUCTURE

3.1 PDRC Physics Specification

- **Mechanism:** Passive Daytime Radiative Cooling (PDRC) targeting the 8–13 µm atmospheric transmission window
- **Metamaterial Metrics:** Solar reflectance $R_{solar} \geq 0.96$ | LWIR emissivity $\varepsilon \geq 0.95$ | SRI ≥ 110 | Net radiative flux 80–110 W/m²
- **Performance:** 5–10°C sub-ambient temperature depression; 20–30% peak electrical cooling demand reduction (ASHRAE 90.1 envelope modeling)

- **Energy Ghost Capture:** Directly quantifies and monetizes 8–13 µm radiative infrared surface heat loss — treated as a recoverable waste asset

3.2 Hardware Product Line

Form Factor	Application	Manufacturer Sourcing Target
Sovereign Skin Wraps	Industrial building envelopes, roofing membranes	Roll-to-roll converters (BOPP, PET, PVDF) — 50–150 µm thickness
High-Solids Elastomeric Spray	Irregular surfaces, HVAC enclosures, vehicle fleets	Industrial coatings toll blenders — min. 65% solids, ≤1.5 lb/gal VOC
Window Wraps / Tints	Commercial glazing thermal reduction	Film laminators with spectrally selective coating capability
Server Rack Skins	Data center thermal management	Precision die-cut converters
GRIPSY Mats	Industrial yards, logistics floors, piezo kinetic harvesting	Vulcanized rubber / polyurethane molders — 80-grit SiC abrasive surface spec

Vendor Sourcing Status: Active outreach via Ground Team Vendor Sourcing Rep. Appendix C outreach script deployed. Qualification questions standardized per hardware vertical.

SECTION 4 — SPEXTER TELEMETRY & HEDERA HCS GATE

4.1 Two-Tier Confidence Ladder

Stage	Method	Confidence	Status
Stage 1	Orbital multi-spectral ghost plume detection	~60% baseline	Diagnostic candidate only — no report issued
Stage 2	Enterprise Specialist physical ground-truth audit	≥95%	Certified — eligible for HCS anchoring and report issuance

4.2 Hedera HCS Confirmation Gate (Mandatory)

GATE RULE: SHA-256 telemetry snapshot MUST be broadcast to the Hedera Consensus Service (HCS) topic and the Transaction ID must confirm on-chain PRIOR to issuing any client-facing forensic report or initiating Solana Token-2022 settlement.

This gate is non-bypassable. Any attempt to issue a report or settle tokens without a confirmed HCS Transaction ID triggers Protocol EXITZ halt.

4.3 Sentinel Gate Triad

Gate	Threshold	Method
Document Integrity	≥ 95%	OCR parsing of uploaded utility bills, invoices, scale receipts
LLM Semantic Confidence	≥ 75%	Cross-reference against ASHRAE 90.1, EPA eGRID, ICE v4.1, Ecoinvent v3.10.1
Spatial / GIS Correlation	≥ 75%	Haversine distance validation, facility footprint cross-reference
Composite Floor	**≥ 72%**	Canonical Design Invariant — unified with On Point notarization floor

4.4 Sentinel Verdict Matrix (7 Tiers)

Verdict	Meaning
AUTO_APPROVED	All gates cleared; settlement proceeds automatically
PENDING_SOVEREIGN_REVIEW	Composite score 68–71%; operator review required
PENDING_AUTHORIZATION	

	Documents received; awaiting counterparty signature or additional uploads
SOVEREIGN_HOLD	EXITZ circuit break active; sole release: Jimi Austin multi-sig key
ESCROW_REVIEW	Settlement funds staged; legal review triggered
PENDING_PLATFORM_REVIEW	Platform-level anomaly flagged; engineering review required
REJECTED	Gates failed; counterparty must resubmit with corrected documentation

SECTION 5 — PROTOCOL EXITZ & UNIVERSAL LAW SETTLEMENT

- **Classification:** Server-side non-monetized circuit breaker — zero token supply, zero public UI exposure
- **Trigger Conditions:** BPS integrity assertion failure, HCS anchor mismatch, On Point anomaly storm, or Sentinel score collapse
- **Action:** Immediate halt of mint/settlement; affected node moves to SOVEREIGN_HOLD verdict
- **Release:** Sole authorization — Jimi Austin (Kato / 👑 Zer0) cryptographic signing key
- **Regulatory Export Schemas:** CSRD (ESRS E1), SEC Climate Disclosures, ISSB S2, GHG Protocol Scope 3 — all settlement records formatted for immediate regulatory export

SECTION 6 — ECHO-EX PUBLIC INTERFACE

- **Domain:** echo-ex.com
- **Eligibility:** Strictly ages 16–70 (hard age gate enforced at intake)
- **Mandatory Terminology:** "Participants" only in all public-facing contexts. "Specialist" is reserved exclusively for certified enterprise audit technicians.
- **MLM Guard:** Hard-coded 2-level maximum referral cap. Programmatically enforced. Zero exceptions.
- **Intake Separation:** Echo-Ex individual participant flow is architecturally isolated from the Bedrock FAD corporate/commercial intake funnel. No crossover routing.
- **BPS Participation:** Echo-Ex Participant settlements route through the standard 7,000 BPS / 2,000 BPS / 1,000 BPS immutable split via Bedrock Flow.

SECTION 7 — DECOUPLED BRANCH: MiK (AGENT ARVIN)

- **Classification:** Independent retail POS / telemetry engine — strictly firewalled from SPEXTER blockchain tokenization rails
- **Stack:** Dutchie API, Twilio E.164 webhooks, operational telemetry reporting
- **Boundary Enforcement:** No BPS settlement routing. No Hedera HCS anchoring. No Solana mint interaction. Fully decoupled.

SECTION 8 — CANONICAL CONSTANTS REFERENCE CARD

Constant	Value
BPS Invariant	Σ = 10,000 BPS | 7,000 Asset Sovereign / 2,000 Verification Node / 1,000 Public Resilience Pool
Unified Confidence Floor	≥ 0.72 (72.00%) — Canonical Design Invariant
Sentinel Triad	≥ 95% Doc | ≥ 75% LLM | ≥ 75% Spatial
Retroactive Window	36 months
Scope 3 Sub-Tracks	24 (9 canonical categories + 24th: Fiat Currency & Cross-Border Logistics)
AETHEXER Thermal Window	8–13 µm
Sub-ambient Depression	5–10°C
Peak Cooling Reduction	20–30% (ASHRAE 90.1)
On Point Notarization Floor	≥ 0.72 (canonical parity with Sentinel)
Echo-Ex Age Gate	16–70 years
MLM Referral Cap	2 levels maximum (programmatic)
Solana Mint Address	Eb4mSecyZh6BQ2wm5rbBnB2GNJu8iuEqGq53YHBVJ88V
Host Infrastructure	208.122.8.11 | systemd bedrock-flow | NEXT_DIST_DIR=.build
EXITZ Release Authority	Jimi Austin (Kato / Zer0) — Sole Multi-Sig Key
Standards Grounding	ASHRAE 90.1, EPA eGRID, ICE v4.1, Ecoinvent v3.10.1, IBI/EBC (Biochar), Eurobitume EPD